

## NGUYET MINH TRANG (DIANA TRANG)

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<https://ngtr99.github.io/html/index.html>

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<https://github.com/ngtr99?tab=repositories>

**Aspiring Electrical Engineer** with hands-on experience in power electronics, embedded systems, and circuit design. Proficient in DC-DC converters, PWM control, sensor interfacing, and IoT hardware solutions using ESP32 microcontrollers. Seeking a hardware-focused Electrical Engineering internship to apply circuit design, power conversion, and embedded control skills.

### SKILLS

**Programming Languages:** Python, C++, Java, JavaScript  
**Embedded & Hardware:** ESP32, Arduino, PWM Control, Sensor Integration, MOSFET  
**Software & Tools:** VS Code, GitHub, MATLAB  
**Circuit Design & Tools:** LTSpice, Breadboarding, Circuit Testing & Debugging

### EDUCATION

**Evergreen Valley College** – *Electrical Engineering*

GPA: 3.8

Expected: December 2026

### PROJECT

**Buck – Boost Converter with ESP32 Control | ESP32, PWM, Power Electronics, Circuit Design, LTSpice**

Apr 2026 – Present

<https://buck-boost-converters.vercel.app/>

- Designed and implemented a DC-DC buck-boost converter using an ESP32 microcontroller for closed-loop PWM control of MOSFET switching.
- Developed high-frequency PWM signal generation to achieve both step-up and step-down voltage conversion, validating the theoretical relationship between duty cycle and output voltage.
- Performed circuit simulation in LTSpice and hardware validation, successfully demonstrating stable voltage regulation across varying load conditions.
- Analyzed system performance, identifying and mitigating switching noise and output ripple through component selection and filtering techniques.
- Successfully achieved expected output voltage regulation in both buck and boost modes, accurately following duty cycle adjustments with stable performance.

**IoT Temperature & Humidity Monitoring System | ESP32, DHT11, IoT, Python, Machine Learning, Full-Stack Development**

Apr 2026 – Present

<https://temperature-humidity-sensor.vercel.app/>

- Developed a complete IoT environmental monitoring system using ESP32 and DHT11 sensor for real-time temperature and humidity data acquisition via Arduino framework (C++).
- Built an end-to-end data pipeline from sensor → microcontroller → cloud → web dashboard, deployed on Vercel.
- Designed and implemented a responsive web interface (HTML, CSS, JavaScript) for live data visualization and system monitoring.
- Utilized Python for data preprocessing, analysis, and basic machine learning models to predict temperature and humidity trends.
- Integrated I2C communication with SSD1306 OLED display for local real-time status and Serial communication for debugging.
- Successfully captured real-time temperature and humidity readings, generated predictions, and displayed both current values and forecasts on the SSD1306 OLED screen.

**Smart Home Automation System | Embedded Systems, Sensor Integration**

Apr 2026 – Present

<https://smart-house-plum.vercel.app/>

- Designed and built an ESP32-based smart home automation system that automatically controls environmental devices based on sensor inputs.
- Integrated DHT11 temperature sensor, HC-SR501 motion sensor, relays, and RGB LED with decision-making logic to automate fan control and lighting.
- Implemented I2C protocol for real-time data display on SSD1306 OLED and user feedback.
- Developed responsive control logic combining sensor data and manual user commands for improved home comfort and energy efficiency.
- Prototyped the full system on a breadboard with proper power management and signal integrity considerations.
- Achieved responsive environmental control: system reliably turns on the fan based on temperature and motion, with real-time status and color-coded feedback displayed on the OLED screen.

### Soft Skills

Problem-solving, teamwork, attention to detail, and time management